Tutorial Topic 1
Functions

Question 1

Given $f(x) = 5 - 2x$ and $g(x) = \frac{2}{x+2}$, evaluate $\frac{f(-2)-g(5)}{3}$.

Question 2

Given $f(x) = \frac{x+3}{1-2x}$ and $g(x) = \frac{1}{x+5}$, evaluate $\frac{2f(-1)+g(-3)}{2}$.

Question 3

Determine the domain of the following functions:

(a) $f(x) = 1 - 2x$

(b) $f(x) = \frac{x}{1-2x}$

(c) $f: x \to \frac{x}{\sqrt{x+1}}$

Question 4

Function $f(x)$ is defined as $f(x) = \frac{36}{a+bx}$, $x \neq -\frac{a}{b}$. If $f(9) = 12$ and $f(8) = 9$,

(a) find the value of a and b.

(b) the value of x which is mapped to 4.

(c) the value that remains the same under the mapping that is $f(x) = x$.

Question 5

Solve for x:

(a) $e^{2x} = 3e^{x}$

(b) $e^{2x} - 7e^{x} + 12 = 0$

(c) $3e^{x} - 5 = -2e^{-x}$

Question 6

Find x for :

(a) $\log_3\left(\frac{1}{9}\right) = x.$

(b) $5^{x+3} = 100$.

(c) $\log_3 x^2 = 1 + \log_3(x + 6).$

(d) $5\log_x 6 - \log_x 96 = 4.$

(e) $7^{x+1} = 2^{3x-2}.$

Question 7

An efficiency study at a certain factory showed that on average, a worker who arrives at work at 8 am can assemble $f(x) = -x^3 + 6x^2 + 15x$ television sets x hours later. How many sets will such a worker have assembled by 10 am?

Question 8

The conversion formula for temperature is $f(x) = 1.8x + 32$ where x is the temperature in degrees Celsius (˚C) and $f(x)$ is in degrees Fahrenheit (˚F).

(a) What are the freezing and boiling temperatures in ˚F ?

(b) Aluminum melts at 1220˚F. What is aluminum's melting point in ˚C?

Question 9

If an object is thrown vertically upward from the ground with an initial speed of 160 ft/s, its height t secs later is given by $H(t) = -16t^2 + 160t$.

(a) Graph the function $H(t)$.

(b) Determine when the object will hit the ground.

(c) Estimate how high the object will rise.

Question 10

A radioactive substance is decaying according to the formula $y = Ae^{kx}$ where x is the time in years. The initial amount is 10 grams and after 5 years, 8 grams remain.

(a) Find k. Leave the answer in terms of natural log.

(b) Estimate the amount remaining after 10 years.

(c) Find the half-life to the nearest tenth of a year.